Smart Health Monitoring Band Using IoT and Machine Learning

Hrishikesh Anand
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India

***Abstract*—Low-cost wrist monitors usually compare each heart-rate reading with one population limit, which misses gradual drifts and low heart rate, and they lose data when Wi-Fi drops. We present an ESP32-based smart band with a MAX30102 photoplethysmography sensor and an MQ135 breath/air sensor that learns each wearer's usual heart rate and air level on the device in two minutes. The baseline is then updated only by windows the band itself judges normal, so an abnormal episode can never become the new normal. Every 30-second window is scored on the band by a hybrid rule combining a decision tree on baseline-relative features, a personal z-score and hard safety limits; the logic is exported to C and matched the training model on 3,754 windows with no mismatch. Windows are queued for up to one hour during network loss and uploaded to a Firebase dashboard where caregivers label alerts for re-training. On 16 simulated wearers unseen during training, the method caught 99 % of harmful events against 48 % for the band's original fixed threshold and raised recall from 0.26 to 0.82, at a false-alarm rate of 2.6 % of normal windows. Exercise remains the main source of false alarms.**

***Index Terms*—wearable health monitoring, photoplethysmography, personalised baseline, edge machine learning, decision tree, ESP32, Internet of Things.**

I. Introduction

Continuous heart-rate monitoring on the wrist has become inexpensive: a photoplethysmography (PPG) sensor and a Wi-Fi microcontroller cost only a few hundred rupees together. Turning those readings into useful alerts is harder. Most low-cost monitors compare each reading with a single population limit, for example 120 BPM. Such a limit misses a wearer whose resting rate is 62 BPM and who climbs steadily to 110 BPM, and it misses slow-onset low heart rate entirely. Clinical scores such as NEWS2 [1] also rely on population bands, and monitors that alarm too often are known to cause alarm fatigue [2].

A natural improvement is to judge each reading against the wearer's own usual level. This raises a problem that is easy to overlook: if the personal reference keeps learning from every reading, an abnormal episode is gradually absorbed into it, and the next episode looks less unusual. A second practical problem is connectivity. Many student and maker designs [5], [6] stop recording when Wi-Fi drops, so the episode that mattered may never reach the caregiver.

This paper presents a wrist-worn smart band built from an ESP32-WROOM-32E, a MAX30102 PPG sensor, an MQ135 gas sensor, an OLED display and a buzzer, together with a cloud dashboard. Its contributions are:

- an **event-gated personal baseline** learned on the band in two minutes and afterwards updated only by windows the band itself classifies as normal;
- a **hybrid on-band status rule** that combines a decision-tree risk score on baseline-relative features, a personal z-score and hard safety limits, exported to C and verified to match the training model exactly;
- a **store-and-forward cloud path** that keeps up to one hour of windows on the band during network loss, and a dashboard whose *Real*/*False alarm* labels flow back into training;
- a subject-independent evaluation against the band's original fixed-threshold firmware, including the cases where the method does not work.

Section II reviews related work, Section III describes the system, Section IV the evaluation method, Section V the results, and Sections VI–VII discuss limitations and conclude.

II. Related Work

A. Low-Cost IoT Health Monitors

Saw [5] uses an Arduino with an ESP8266 to collect vital signs and publish them to a cloud platform; decisions are fixed limits applied after upload. Sapkal's Smart Life Band [6] places a heart-rate/SpO_2 sensor, an IMU, an OLED and an SOS button on an ESP32 and sends WhatsApp alerts from a Firebase-backed web page. Both are useful reference designs, but neither keeps a personal baseline, runs a learned model on the device or retains readings through a network outage.

B. Personal Baselines in Wearable Patents

Several patents compare live data with a personal reference. US20180325385A1 [7] stores baseline profiles for a wearable that senses heart rate, oximetry and environmental particulates, and alerts caregivers when health status deviates from the baseline by a set percentage. US20180296105A1 [8] evaluates heart-rate-variability figures of merit against baselines and limits, and US11410765B2 [10] builds an HRV baseline from several days of data. None of the summaries we reviewed describes keeping flagged episodes out of the baseline, which is the core of this work. Other patents target acute events with wrist PPG [9], population early-warning scores from wearable data [11], offline machine-learning decisions with later upload

in emergency-response devices [12], and risk prediction for heart-failure decompensation [13].

C. Learning at the Edge

Giordano *et al.* [14] show that useful inference fits on an ultra-low-power IoT node, while Scheid *et al.* [15] show that sequences of wearable observations predict in-hospital deterioration better than isolated readings, using clinical hardware and a large server-side model. Our work sits between these: a sequence-aware but very small model that runs on a single low-cost microcontroller.

III. System Design

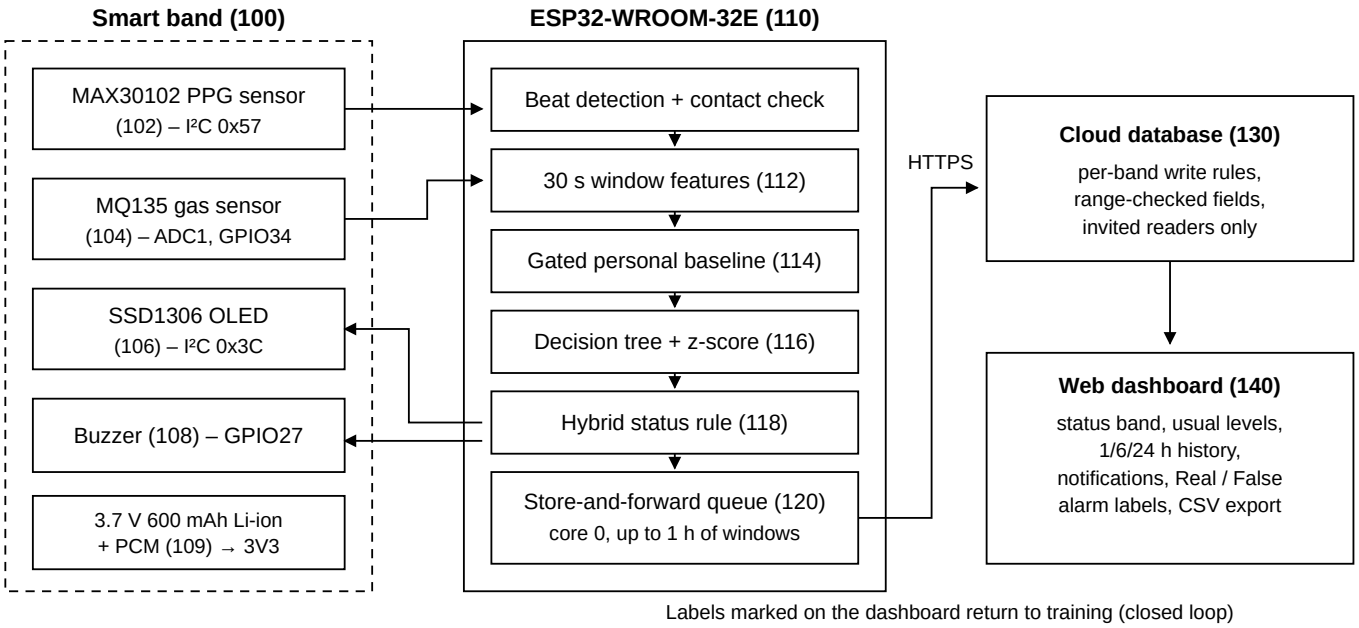


Fig. 1. System architecture of the smart band and its cloud path. Numbers in brackets identify the blocks referred to in the text.

A. Hardware

Fig. 1 shows the architecture, Fig. 2 the prototype and Fig. 3 the wiring. The MAX30102 [18] and an SSD1306 OLED share one I²C bus (SDA GPIO21, SCL GPIO22, 3.3 V). The MQ135 [19] analogue output is read on GPIO34, an ADC1 channel that remains usable while Wi-Fi is active [20]. A buzzer is driven from GPIO27. The unit runs from a 3.7 V, 600 mAh Li-ion cell through a protection module. Everything is housed in a transparent enclosure on a hook-and-loop wrist strap; the OLED faces the wearer and the MQ135 mesh is exposed to air. The MQ135 responds to gases in exhaled breath and the surrounding air, so it is treated as a relative breath/air index rather than a calibrated concentration.



(a) (b)

Fig. 2. Hardware prototype. (a) Front view: the OLED shows heart rate, air index, on-band risk and z value, cloud status and a finger-placement prompt. (b) Side view on the hook-and-loop wrist strap, showing the transparent enclosure and internal wiring.

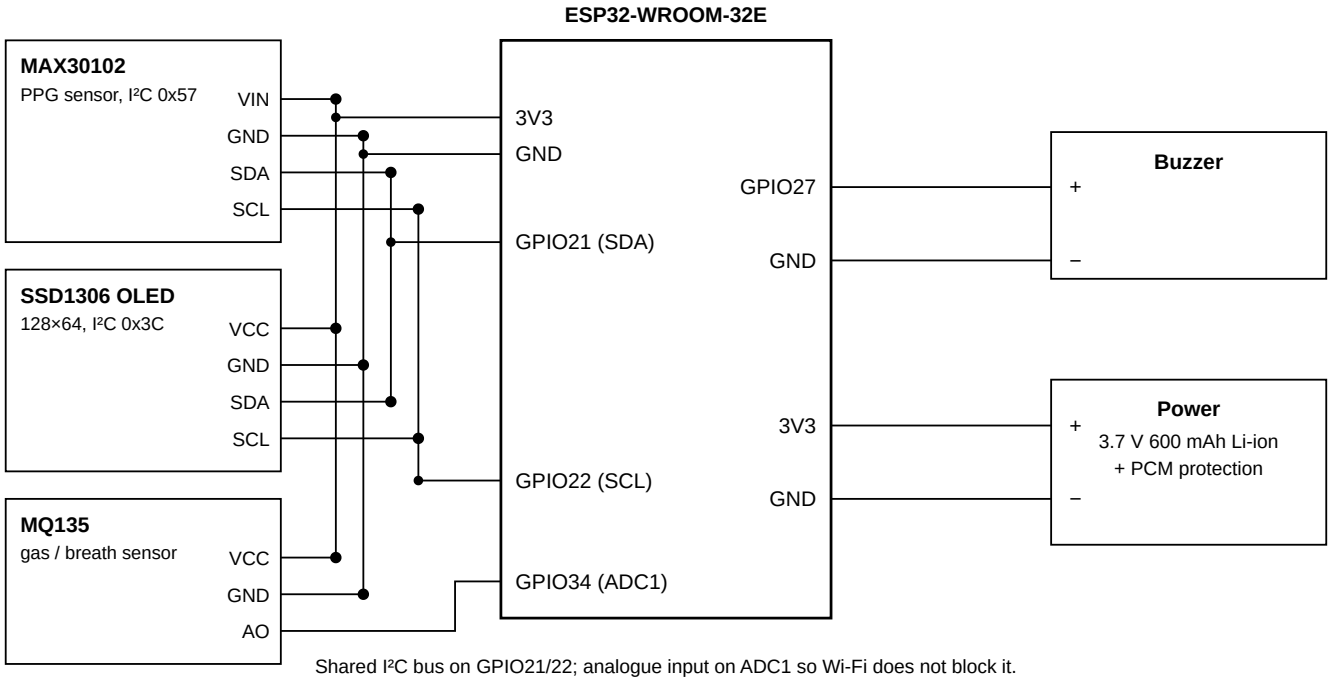


Fig. 3. Wiring diagram. MAX30102 and OLED share the I²C bus; the MQ135 output is read on ADC1 (GPIO34); the buzzer is on GPIO27.

B. Window Features

Beats are detected from the infrared PPG signal [3]. Heart rate is a rolling average of recent beats, accepted only between 40 and 200 BPM. Every 30 s the band closes a window and computes the fraction of time with a valid pulse, the *coverage*. Windows with coverage below 60 % are marked NO CONTACT and are neither scored nor learned from. From each remaining window the band computes nine features: mean, spread and slope of heart rate; mean and slope of the MQ135 value; and, for both signals, the delta and z-score relative to the wearer's baseline.

C. Event-Gated Personal Baseline

The first four valid windows (two minutes) initialise the baseline b and spread σ for heart rate and for the MQ135 value. During this phase the band shows LEARNING. Afterwards, for each signal x in window t,

$$b_t = (1 - \alpha)\, b_{t-1} + \alpha\, x_t \ \text{ if } s_t = \text{NORMAL}, \ \text{else } b_t = b_{t-1} \quad (1)$$

where s_t is the status the band assigns to the window (Section III-D) and α is a small smoothing factor. The baseline therefore follows slow, genuine changes such as time of day, but a window the band has flagged can never pull the "usual" level towards itself. Fig. 4 illustrates the difference on a synthetic trace: a baseline that learns from every window climbs during the episode, while the gated baseline stays at the wearer's usual level. The z-scores used as features are

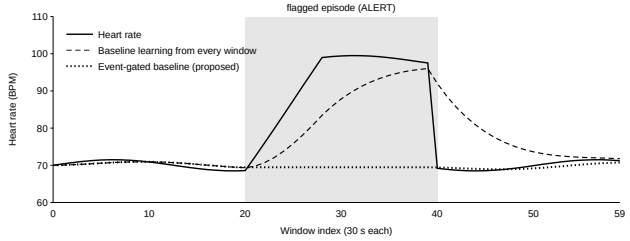


Fig. 4. Illustrative synthetic trace (not measured data): a baseline that learns from every window drifts upward during a flagged episode, while the event-gated baseline does not.

$$z_t = (x_t - b_{t-1}) / \sigma \quad (2)$$

D. Hybrid Status Rule

A decision tree [16] maps the nine features to an alert probability r_t. The anomaly score is

$$a_t = \max(|z_{\text{HR},t}|, |z_{\text{MQ},t}|) \quad (3)$$

A window is ALERT if r_t or a_t reaches its alert cut-off, or if a hard safety limit is crossed (HR ≥ 150 BPM, HR ≤ 40 BPM or MQ135 ≥ 900). Lower cut-offs give WARNING; otherwise the window is NORMAL. Fig. 5 summarises the decision map and Fig. 6 the full per-window procedure. The safety limits act independently of the model, so a model error cannot suppress an obviously dangerous reading. ALERT pulses the buzzer and WARNING gives one chirp per window.

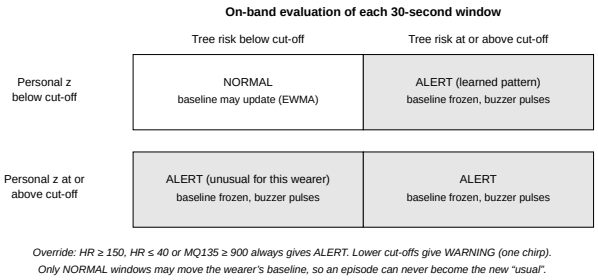


Fig. 5. Decision map of the hybrid status rule and baseline gating.

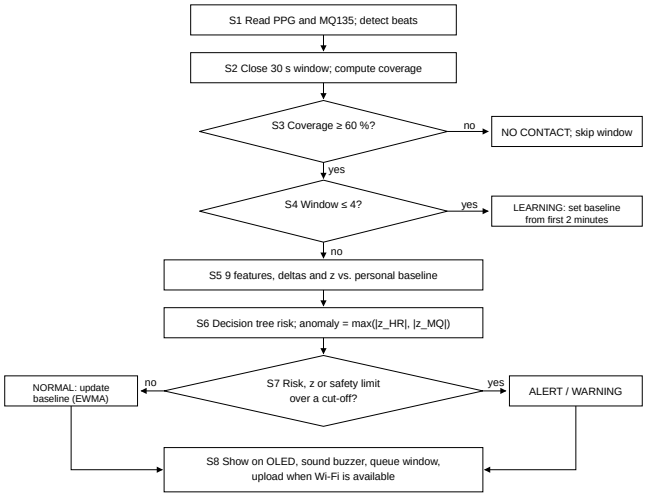


Fig. 6. Per-window procedure on the band, steps S1–S8.

E. Edge Export

The trained tree, cut-offs and baseline logic are generated automatically as a C header (*edge_model.h*), compiled into the firmware and replayed against the Python model on the test windows before deployment. The tree reduces to nested comparisons, so evaluation takes negligible time on the ESP32.

F. Store-and-Forward Upload and Dashboard

Scored windows are placed in a queue and uploaded over HTTPS by a task on the ESP32's second core, so beat detection is never paused by networking. During Wi-Fi loss up to one hour of windows is retained and then uploaded in order; the band verifies the server certificate and shows the cloud state on the OLED (Fig. 7). Windows are stored in a Firebase Realtime Database whose rules [21] let each band write only its own path, let only invited caregivers read, and range-check every field.

The dashboard (Fig. 8) requires sign-in and shows a status band (Normal, Warning, Alert, Learning, No contact or Offline), the live heart rate with a ten-minute trace, the sensor reading, the on-band risk and the "how unusual" z value. History charts over 1, 6 and 24 h plot each signal against the wearer's usual level with alert periods shaded. Each alert episode can be marked *Real* or *False alarm*, and readings export as CSV in the training format, so confirmed episodes become labels for re-training.

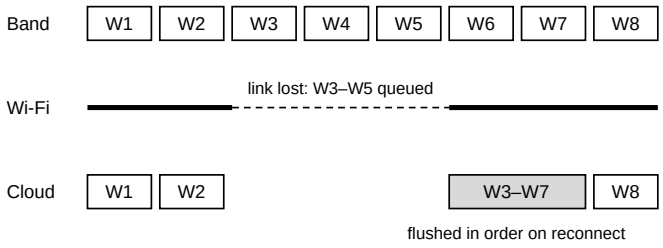
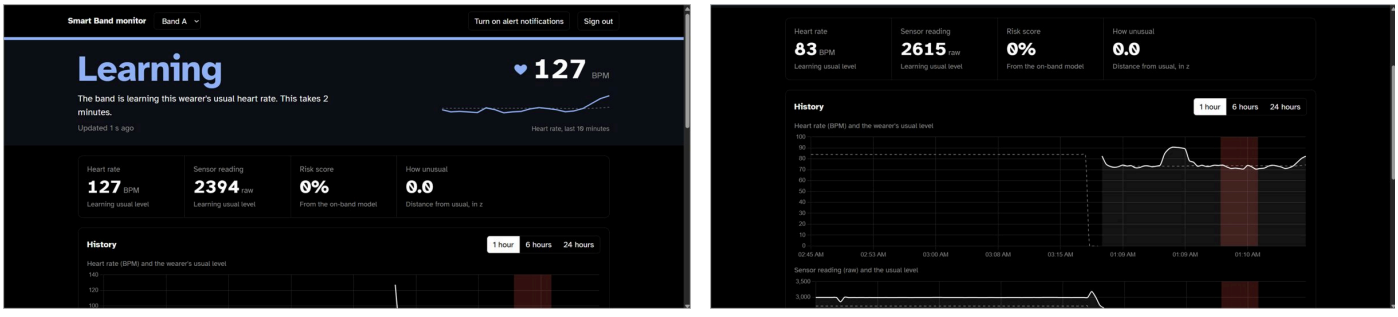


Fig. 7. Store-and-forward upload during a Wi-Fi outage.



(a) (b)

Fig. 8. Web dashboard. (a) Learning phase: status band, live heart rate with usual-level trace, sensor reading, on-band risk and z value. (b) History: heart rate and sensor reading against the wearer's usual level (dashed) with an alert period shaded.

IV. Evaluation Method

A. Data

Real recordings logged so far are too few for statistical evaluation, so detection performance was measured on simulated wearers. Each simulated wearer has a personal resting heart rate and room air level and experiences normal periods, exercise, sudden heart-rate rises, gradual drifts, low heart rate and gas events, with sensor noise. Ground-truth labels mark windows that should raise an alert. A serial logger and the dashboard CSV export are in place to replace the simulation with real sessions.

B. Training and Splitting

Adjacent windows from one person are strongly correlated, so splitting by window would leak each wearer into both training and test sets. We therefore split by wearer: the test set contains 16 wearers never seen during training or tuning. Tree hyper-parameters (depth, leaf size, criterion and class weights) were chosen by a grid of 96 combinations with five-fold cross-validation grouped by wearer, using scikit-learn [17]. Cut-offs for the status rule were chosen on out-of-fold predictions only.

C. Comparison and Metrics

The comparison method is the band's original firmware: ALERT when HR ≥ 120 BPM or MQ135 ≥ 700. We report window-level recall, F1 score and false-alarm rate (the fraction of normal windows that raise ALERT), and event-level detection, the fraction of harmful episodes in which at least one window raises ALERT.

TABLE I
EVALUATION SET-UP

Item	Setting
Window	30 s; coverage < 60 % dropped
Learning phase	First 4 windows (2 min)
Features	9 per window
Model	Decision tree, grid of 96 × 5-fold GroupKFold
Test set	16 unseen simulated wearers
Comparison	HR ≥ 120 or MQ135 ≥ 700

V. RESULTS

A. Detection

Table II and Fig. 9 compare the two methods. The fixed threshold caught fewer than half of the harmful events: it missed every low-heart-rate episode and every rise that stayed below 120 BPM. The proposed rule caught 99 % of events and raised recall from 0.26 to 0.82. The cost is a false-alarm rate of 2.6 % of normal windows against 0.2 %.

TABLE II
DETECTION ON 16 UNSEEN SIMULATED WEARERS

Method	Recall	F1	False alarms	Events caught
Fixed threshold	0.26	0.41	0.2 %	48 %
Proposed	0.82	0.84	2.6 %	99 %

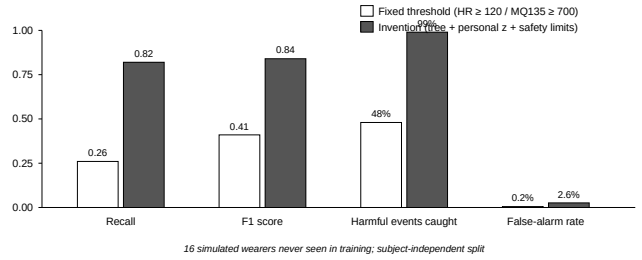


Fig. 9. Fixed threshold versus the proposed hybrid rule on unseen simulated wearers.

B. Early Warning

In five gradual heart-rate rises the proposed rule raised a warning before the ground-truth label became harmful. A single fixed limit cannot do this, because by construction it reacts only once the limit is crossed.

C. Exercise

About 39 % of normal exercise episodes still triggered an alert. Without a motion sensor the band cannot tell a heart-rate rise caused by activity from one caused by illness, and wrist PPG itself degrades during intense movement [4]. This is the main source of false alarms.

D. Ablation of the Anomaly Rule

Removing the separate z-score rule changed detection only slightly, because the tree already uses baseline-relative features. Keeping it costs about 0.5 % more false alarms; we retain it as a safety net for patterns absent from training.

E. Edge Equivalence and Prototype Operation

The generated C model was compiled and replayed against the Python model on 3,754 windows with zero mismatches. On the physical band we observed beat detection, the finger-placement prompt, the two-minute learning phase, on-band risk and z display, buzzer alerts and the "cloud ok" state, and the dashboard received live windows and plotted them against the usual level (Figs. 2 and 8).

TABLE III
SUMMARY OF TEST CASES

Test	Result	Status
Edge vs. Python	3,754 windows, 0 mismatches	Met
Detection	Events 99 % vs. 48 %	Met; more false alarms
Early warning	5 gradual rises	Met
Exercise	~39 % alert	Not met
Anomaly ablation	+0.5 % false alarms	Trade-off
Prototype + cloud	All functions observed	Functional

VI. DISCUSSION AND LIMITATIONS

The gated baseline is what makes personal z-scores trustworthy over time: because flagged windows never update it, a long episode remains unusual for as long as it lasts instead of slowly becoming the new normal. The price is that a genuine, lasting change in the wearer's resting level is learned only through windows judged normal, so after a large permanent shift the band may keep warning until the caregiver confirms it as a false alarm; a re-learning action triggered from the dashboard is planned for this case.

The results have clear limits. All detection figures come from simulated wearers; they show that the pipeline behaves as designed, not that it is accurate on real people. The MQ135 gives a relative index until its baseline resistance is calibrated, and it is not a medical breath analyser. The false-alarm rate is dominated by exercise, which an accelerometer should largely remove. Finally, the band is an academic prototype and not a certified medical device.

VII. Conclusion and Future Work

We presented a low-cost ESP32 smart band that learns each wearer's usual heart rate and air level in two minutes, protects that baseline by learning only from windows it judges normal, and scores every window on the device with a hybrid of a decision tree, a personal z-score and fixed safety limits. On unseen simulated wearers it caught 99 % of harmful events against 48 % for the band's original fixed threshold, gave early warnings on gradual rises, and kept readings through network outages on their way to a secure dashboard. Future work will log real sessions from several volunteers and re-train on them, add an accelerometer for activity context, add SpO_2 features from the MAX30102 red channel, calibrate the MQ135, and measure battery life and upload latency on the physical band.

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